Next-day direction prediction with regime featuers

Rationale

One of my professors once said that the essence of statistics is to unpuzzle complex real-world problems into numbersso that they can be analyzed quantatively. At first, weather forecasting seemed like the most complex problem to me, as it involves physical phenomena that cannot be modeled and even official forecasts are often inaccurate.

However, it did not take long for me to realize that financial markets may be even more difficult to predict. This is because market behavior is influenced by multiple factors, including current events, politics, investor psychology, and many other interacting factors, making financial predictions intellectually interesting but challenging. This motivated me to explore two books on quantitative finance: *Python for Finance* by Yves Hilpisch and *Advances in Financial Machine Learning* by Marcos López de Prado. Although these books provided fascinating conceptual insignts, I found it difficult to fully understand the material without implementing the ideas in code. This project was therefore designed as an introductory attempt to predict the next-day direction of a stock index while incorporating key ideas from quantitative finance, such as feature selection, model evaluation, and backtesting.

Problem Definition

The primary objective of the project is to predict the direction of a next-day market. There are various features regarding the market that the model could predict, but this project will focus on the daily close price, as it is the most widely used variable as a label for training. Since the model estimates the daily close price, the features would include the daily prices, and various manipulations of the such, including the volatility, moving average, etc.

Data Description

There are countless numbers of stock samples that this project could refer to train and evaluate the model. Especially, there are a few models in which the model could perform effectively by overfitting, but this does not reflect that the model was designed effectively. Hence, in order to prevent overfitting, the model is evaluated on the S&P 500 index (^GSPC), one of the most widely followed benchmark indices in financial markets. Moreover, as a source of market regime, the model refers to the VIX index (^VIX), an index that outlines the market's expectation of future volatility derived from S&P 500 options.

Deciding the periods for training and testing was not intuitive. Although the members of the project had previous experiences in machine learning projects such as weather forecasting projects included chronological order of training and testing, i.e., using past data to predict the future, but for financial machine learning, many examples including the ones from *Advances in*

Financial Machine Learning by Lopez introduced examples that divides the time series into three sections, where the chronologically first and the last set is used to train the model, and the second slot is used to test the model. Despite the advantage, this adds additional complexity to the problem, as it imposes future-leakage problems, and this project wanted to leave this as a coursework for future projects, as this leads to various different techniques including k-purged folding.

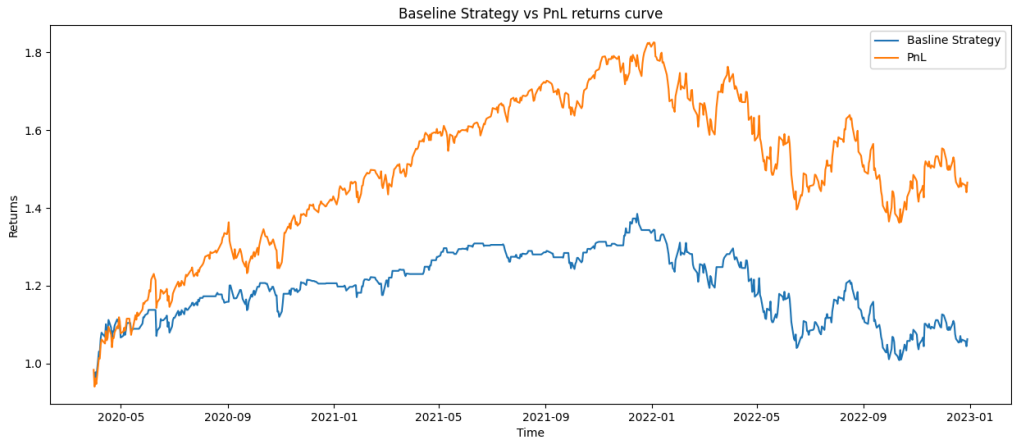
Features and Labels

The model includes several standard features that are widely used in financial machine learning, including lagged returns, trend features, volatility regime, mean-reversion indicators, and regime variables. More specifically, the model includes up to five lagged close-based values as features. The number was determined after experimenting with several length lags, although this process was not systematic and should not be considered as an optimal feature-selection procedure. Similarly, the lookback windows used for features such as momentum and z-score were chosen through experiments. Thus, feature engineering remains as one of the most important areas for improvement in this project.

Modeling Framework

Python for Finance by Yves Hilpisch introduces various modeling frameworks, including random forests, support vector machines, and other supervised learning approaches. Despite the options, due to lack of experience, it was difficult for me to determine the optimal modeling framework. In this project, a random forest classifier was chosen because it is a relatively accessible and well-established model, and because it was presented in the book as one of the fundamental machine learning methods for financial applications. While this choice was reasonable for this introductory project, model selection remains another area for improvement in future work.

Backtesting Methodology



After constructing the baseline strategy, its performance was evaluated using several trading metrics, including the Sharpe ratio, compound annual growth rate (CAGR), yearly turnover, hit rate, and maximum drawdown.

The baseline results are shown as follows:

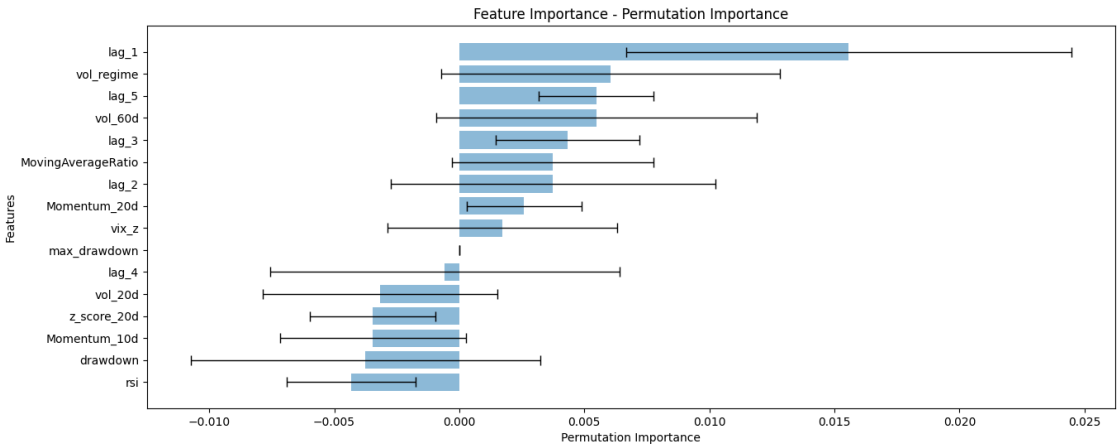
- 2020 Sharpe: 1.2138
- 2021 Sharpe: 1.0833
- 2022 Sharpe: -0.9688
- CAGR: 0.0393
- Yearly Turnover: 72.7273
- Hit Rate: 0.4916
- Max Drawdown: -0.2284

The model performed relatively well during the bullish market conditions of 2020~2021, while the performance shows a significant decrease in bearish market conditions of 2022. This suggests that the model may capture short-term momentum, which is more effective in upward trending markets. Although the strategy produces positive returns during certain periods, the compound annual growth rate remains modest at approximately 3.93%.

The strategy exhibits a relatively high yearly turnover of 72.73. However, this is because this study focuses mainly on evaluating the theoretical predictive capability rather than practical implementability. Due to this reason, transaction costs and slippage was not incorporated in this stage. If practical implementability was considered, the model threshold must be re-evaluated to control the yearly turnover.

Yet, the model has notable weaknesses. The hit rate is approximately 49%, which is close to random guessing. Although profitability in trading strategies does not neecessarily relate to high hit rates, this clearly suggests that the performance could be enhanced. Moreover, the model exhibits a high maximum drawdown of -22.8% which reveals its weaknesses in downside risks, suggesting risk-management mechanisms must be developed.

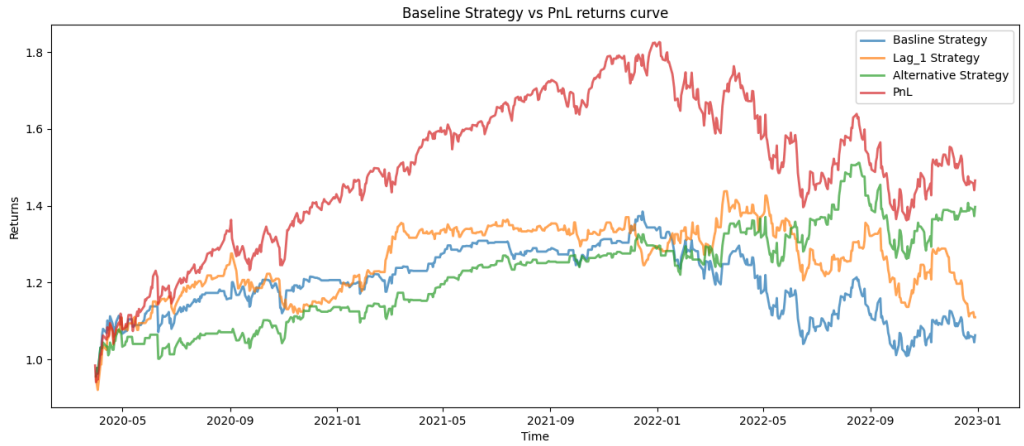
Feature Engineering



To further improve the model performance, permutation importance of the features were analyzed using the *sklearn* library. The feature-importance analysis suggests that short-term lagged features and volatility regime variables contribute the most to the model predictions. On the other hand, technical indicators including RSI, drawdown, and z-score show negative importance values, portraying that they may introduce noise instead of predictive information.

Based on the observation, two alternative models were developed as a comparison. The first model uses only the influential feature lag_1, while the second model considers only the features with the positive permutation importance values.

Alternative Model Development



Statistics for Lag_1 model:

- 2020 Sharpe: 1.0537
- 2021 Sharpe: 0.916
- 2022 Sharpe: -0.745
- CAGR: 0.0557
- Yearly Turnover: 206.5455
- Hit Rate: 0.5225
- Max Drawdown: -0.2284

The Lag_1 model achieved a slightly higher hit rate of 52.3%. However, it also exhibits a high yearly turnover. In other words, the model has some abilithy predicting the direction of next-day returns, but lacks its ability in predicting the return magnitude. This is an unsurprising result, as the model relies on a single lag-based feature. Nevertheless, it is still notable that this model has a higher cummulative annual growth rate compared to the baseline strategy.

Statistics for Alternative model:

- 2020 Sharpe: 0.9401
- 2021 Sharpe: 1.4214
- 2022 Sharpe: 0.3785
- CAGR: 0.1479
- Yearly Turnover: 98.9091
- Hit Rate: 0.5161
- Max Drawdown: -0.1655

The alternative model displays better results compared to the baseline strategy in most metrics, including CAGR, hit rate, and maximum drawdown, and more. Most notably, the model maintained positive Sharpe ratios for all three years. In other words, the model remained profitable on a risk-adjusted basis during the bearish market regime of 2022, suggesting that the revised feature set may provide better robustness to regime shifts compared to both the baseline strategy and lag-based model.

Overall Interpretation

The baseline strategy served as a meaningful benchmark, but its limitations were clear. Since its feature set was based on commonly used indicators applied to S&P 500, rather than on a systematically justifed feature selection process, its performance was not particularly surprising.

The Lag_1 model provided an informative comparison, but its turnover was far too high for real-life implementation. While it performed reasonably well in experimental simulation, it would likely become ineffective once transaction costs were taken into account. In addition, its performance was probably helped by the bullish market regime of 2020-2021. If the testing sample contained a greater proportion of bearish periods, its returns may have deteriorated noticeably.

Among the approaches, the alternative model produced the strongest results. Althrough its turnover remained high, this could potentially be reduced through adjusting the threshold. Importantly, the model remained profitable even during the bearish market condition of 2022, which suggests stronger robustness that the other approaches. However, because the model was tuned using the same asset and sample period, the model has overfitting problems. As a result, this model may not generalize well to other stock and market environments. Potential future would include training and evaluating the framework across a broader range of assets.

Revised Future Work

This project was challenging as I had limited prior experience in designing quantative trading models. In particular, several decisions which were difficult to make, including which features to include, how to determine the threshold values, how to splied the training and testing periods, and which assets to use for model development. This experience made it clear that I need to gain more practical experience, either by working on additional projects or by collborating with others.

In future work, I would like to build a more advanced project incorporating ideas from Lopez de Prado, such as purged k-fold cross validation, meta-labeling, and triple-barrier labeling. These methods may help improve both the rigor and robustness of the overall framework.